

Pipeline leak detection using acoustic frequency analysis

Team: DIMO



- Over 1 billion tons of water are lost annually due to aging pipelines in South Korea.
- Major economic losses and raising concerns about urban infrastructure safety.
- Many buildings that have reached the point of aging.

Pipeline leak detection methods

Smart Ball



Lack of consistency
Lack of prevention
Lack of practicality

Leak Noise Correlator



Smart Double-Lock Pipe



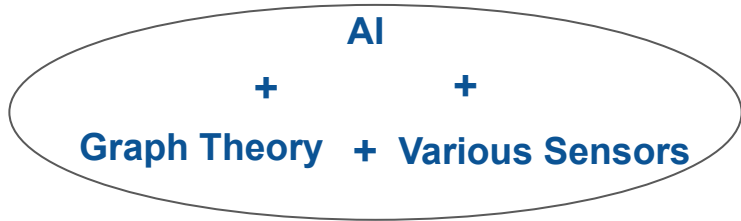
스마트 더블락 누수감지
폴리에틸렌 파복강관



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Introduction

1) Technical Goal



Lack of consistency
Lack of prevention
Lack of practicality



fulfillment consistency
fulfillment prevention
fulfillment practicality

2) Environmental Goal

SDG 13: Climate Action

- Reduces energy use and carbon emissions in water purification and distribution.

SDG 6: Clean Water and Sanitation

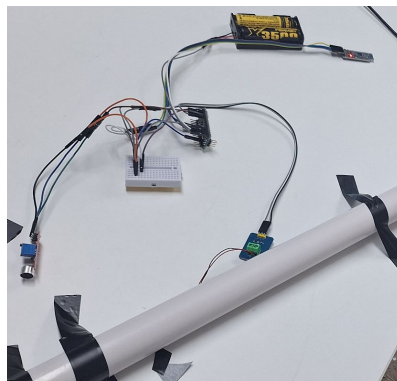
- Minimizes water loss and enhances drought resilience.
- Strengthens stable and equitable water supply for all regions.

SDG 11: Sustainable Cities and Communities

- Prevents secondary damages such as ground subsidence, soil contamination, and road collapse.

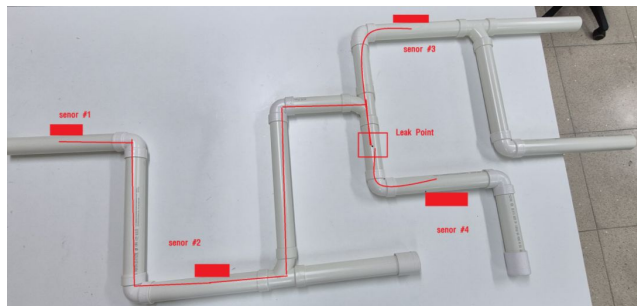


Materials & Methods



leak point

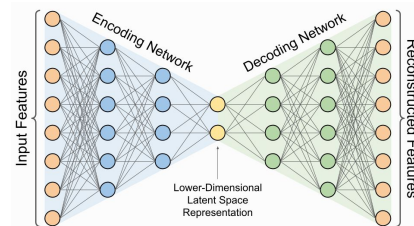
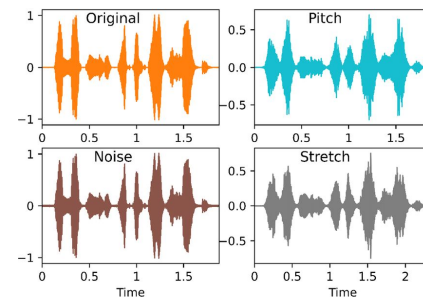
Comparison between ultrasonic sensors and vibration sensors



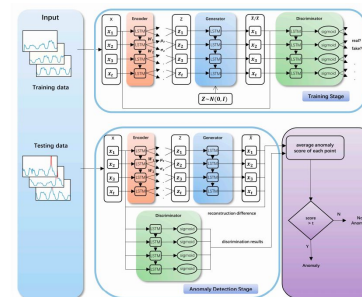
This is a process of estimating the leak location by analyzing sounds according to the distance from the leak, using five different deep learning models for distance measurement.

Data Augmentation

- Time Stretching
- Pitch Shifting
- Volume Augmentation
- Speed perturbation
- Noise Injection

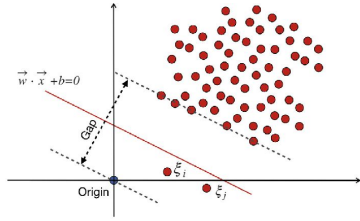


Autoencoder



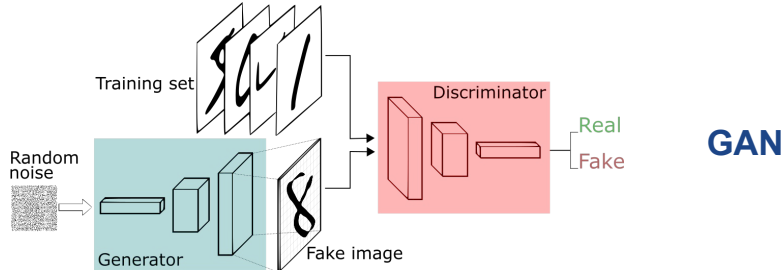
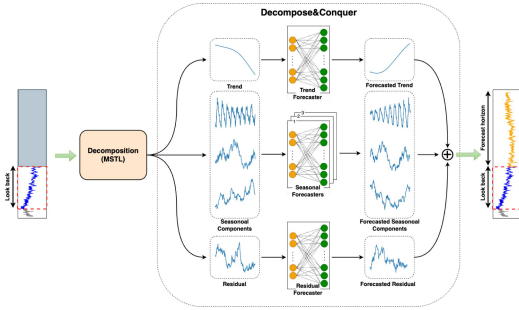
LSTM-VAE

Materials & Methods



One-Class SVM

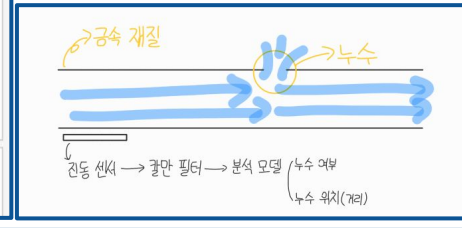
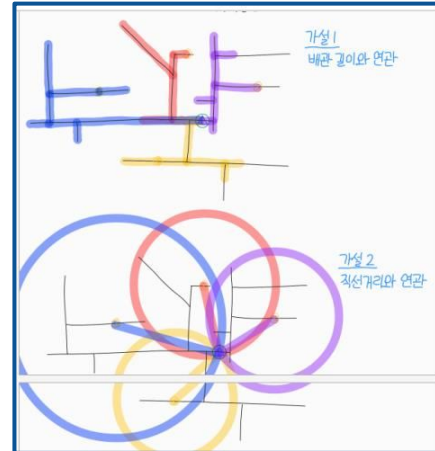
STL Decomposition



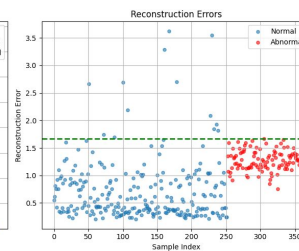
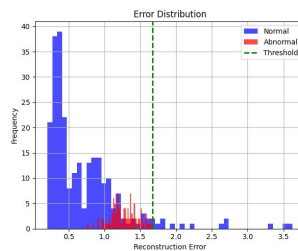
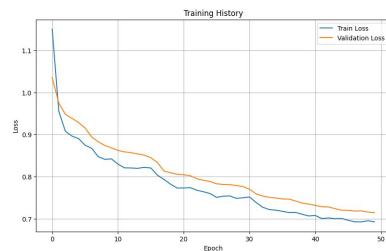
Leak location identification

Hypothesis 1: The sound and ultrasonic sensor readings from an installed leak detection device will be inversely proportional to the total length of the pipe path from the device to the leak location.

Hypothesis 2: The sound and ultrasonic sensor readings from an installed leak detection device will be inversely proportional to the linear distance from the device to the leak location.

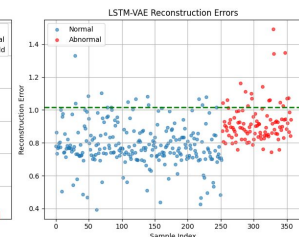
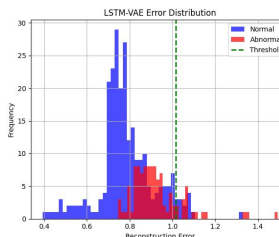
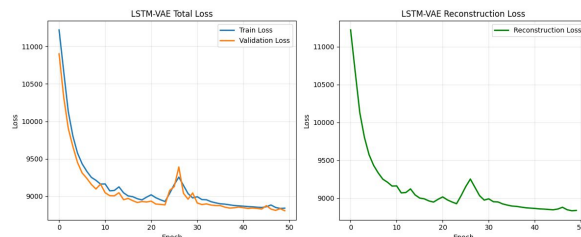


Results



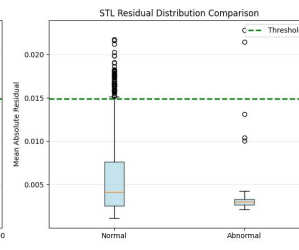
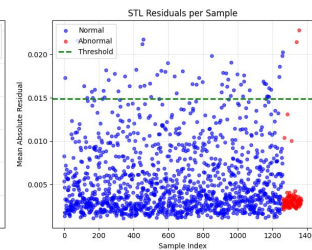
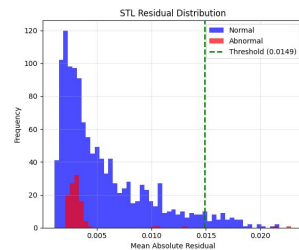
Acc: 0.6527
Precision: 0.5938
Recall: 0.1810
F1: 0.2774

Autoencoder



Acc: 0.7319
Precision: 0.2778
Recall: 0.0476
F1: 0.0813

LSTM-VAE



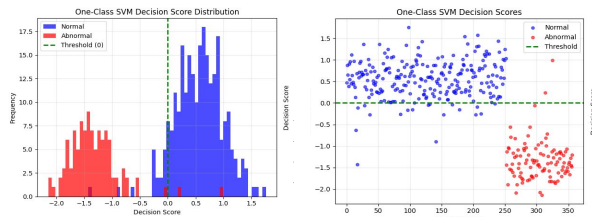
Acc: 0.8184 / 0.5724
Precision: 0.0308
Recall: 0.0190
F1: 0.0235

STL-Decomposition

Results

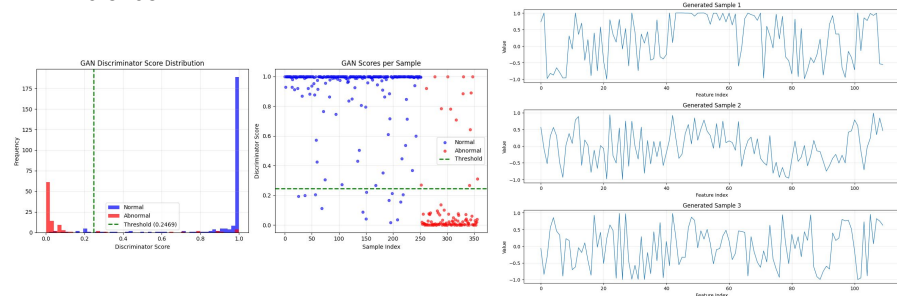
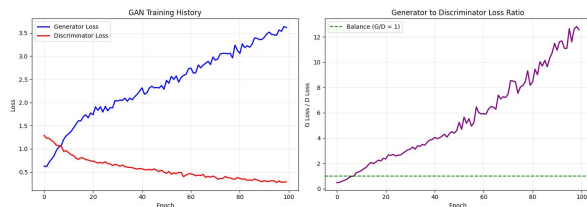
One Class SVM

Acc: 0.4221
Precision: 0.8512
Recall: 0.9810
F1: 0.9115



GAN

Acc: 0.9244/0.5812
Precision: 0.8750
Recall: 0.8667
F1: 0.8708



Conclusion & Future Directions

Conclusion

Autoencoder showed 65% accuracy with fast learning but poor temporal modeling. **LSTM-VAE** captured time dependencies effectively and achieved the highest accuracy of **73%**. **GAN** was slow and less accurate (**58%**), and **One-Class SVM** had large errors (**42%**). **STL-Decomposition** performed well without noise (**81%**) but dropped under noisy conditions (**57%**).

Increasing the number of sensors reduced the error range (**15 cm** → **10 cm** → **5 cm**) and improved **practicality, prevention, and sustainability**.

Future Directions

- Adoption of advanced signal processing and deep learning techniques: Applying a wavelet transform could help reduce noise and enhance the accuracy of analysis, exploring deep learning models for more robust feature extraction and anomaly detection
- Real-time implementation and system integration: Optimize data preprocessing and leak point locating algorithms for real-time operation on embedded DSP devices or edge computing nodes.